

Exercise 1: Control Structures

Scenario 1: The bank wants to apply a discount to loan interest rates for customers above 60 years old.

Question: Write a PL/SQL block that loops through all customers, checks their age, and if they are above 60, apply a 1% discount to their current loan interest rates.

Program

```
DECLARE
```

```
    v_age NUMBER;
```

```
BEGIN
```

```
    FOR cust IN (
```

```
        SELECT CustomerID, DOB
```

```
        FROM Customers
```

```
    )
```

```
    LOOP
```

```
        v_age := FLOOR(MONTHS_BETWEEN(SYSDATE, cust.DOB) / 12);
```

```
        IF v_age > 60 THEN
```

```
            UPDATE Loans
```

```
            SET InterestRate = InterestRate - 1
```

```
            WHERE CustomerID = cust.CustomerID;
```

```
            DBMS_OUTPUT.PUT_LINE('Updated loan interest for Customer ID: ' || cust.CustomerID);
```

```
        END IF;
```

```
    END LOOP;
```

```
    COMMIT;
```

```
EXCEPTION
```

```
    WHEN OTHERS THEN
```

```
        ROLLBACK;
```

```
DBMS_OUTPUT.PUT_LINE('Error: ' || SQLERRM);  
  
END;
```

Scenario 2: A customer can be promoted to VIP status based on their balance.

Question: Write a PL/SQL block that iterates through all customers and sets a flag IsVIP to TRUE for those with a balance over \$10,000.

Program

```
ALTER TABLE Customers  
  
ADD IsVIP CHAR(1);  
  
BEGIN  
  
    FOR cust IN (  
  
        SELECT CustomerID, Balance  
  
        FROM Customers  
  
    )  
  
    LOOP  
  
        IF cust.Balance > 10000 THEN  
  
            UPDATE Customers  
  
            SET IsVIP = 'Y'  
  
            WHERE CustomerID = cust.CustomerID;  
  
            DBMS_OUTPUT.PUT_LINE('Customer ' || cust.CustomerID || ' promoted to VIP.');  
        ELSE  
  
            UPDATE Customers  
  
            SET IsVIP = 'N'  
  
            WHERE CustomerID = cust.CustomerID;
```

```
        END IF;

    END LOOP;

    COMMIT;

EXCEPTION

    WHEN OTHERS THEN

        ROLLBACK;

        DBMS_OUTPUT.PUT_LINE('Error: ' || SQLERRM);

END;
```

Scenario 3: The bank wants to send reminders to customers whose loans are due within the next 30 days.

Question: Write a PL/SQL block that fetches all loans due in the next 30 days and prints a reminder message for each customer.

Program

```
BEGIN

    FOR loan_rec IN (

        SELECT c.Name,

               l.EndDate

        FROM Customers c

        JOIN Loans l

        ON c.CustomerID = l.CustomerID

        WHERE l.EndDate BETWEEN SYSDATE AND SYSDATE + 30

    )

    LOOP

        DBMS_OUTPUT.PUT_LINE(
```

'Reminder: Dear ' || loan_rec.Name ||

', your loan is due on ' ||

TO_CHAR(loan_rec.EndDate, 'DD-MON-YYYY')

);

END LOOP;

END;